Python Freezing Analysis 1.5

NOTE 1: These instructions are for Windows with Anaconda installed

NOTE 2: This code only works with the following video extensions: .mov, .wmv, .mp4

NOTE 3: If using the code for the first time, follow the "Installing Required Libraries" and "Setting Parameters" section first

1. Organise the data folders and the .py file as shown below:
    a. No folders other than ones containing data to be analysed should be in the same folder as the Python script
    b. Naming of data folders does not matter

Name	Date modified	Type	Size
1ul_bio	8/1/2025 10:35 AM	File folder	
1ul_water	8/1/2025 10:38 AM	File folder	
2ul_bio	8/1/2025 10:35 AM	File folder	
2ul_water	8/1/2025 10:35 AM	File folder	
freezing_analysis_1.5.py	8/14/2025 12:12 PM	PY File	15 KB

2. Each data folder should have only two files: the Coldstage TSV file containing temperature data and the video file
    a. Other files present won't affect the code, but it's best to leave the folders as least cluttered as possible, since the plots will be generated in the same folder as the data.

3. Open the "Spyder" app.
    a. Go to the search bar at the bottom left of Windows and type in "Spyder" or open Spyder through Anaconda Navigator
    b. In Spyder, go to the top left menu under "File" and select "Open"
    c. Find and open the Freezing Analysis Code

        i. Make sure the code selected is in the right folder (the folder with the data that needs to be analysed)

4. To begin analysis, click anywhere on the left window with the code and press “Ctrl + Enter” or press “Run” on the top menu bar
    a. The code should then print the location of the file in the console (bottom right window)
    b. The code will cycle through each data folder in the same folder as the code
        i. The code will ask whether to analyse each folder. Type “Y” to continue or “N” to skip to the next folder. Lowercase is okay. Typing anything other than “Y” will be treated as "N"
    c. The code will ask for the number of plates in the trial. Input a positive, non-zero integer
    d. The code will output the number of droplets detected and ask whether manual detection is needed
        i. Answering “Y” opens the detected_droplets image in an interactive window
        ii. Left click to add droplets, right click to remove them. Droplet size of manual droplets is based on the MIN_RADIUS parameter
        iii. Close the window when finished
    e. The code will prompt for the Video Time, File Time, and Skip Seconds for each detected folder in the console
        i. Make sure to input these times correctly and press “Enter”
        ii. The input cannot be changed after pressing “Enter”
        iii. To stop and rerun the code, press the red square at the top right of the console and start again from Step 4

```
--- Detected folder: 1ul_bio_redo ---
Analyze folder '1ul_bio_redo'? (Y/N): y
Video: WIN_20250724_1ul_bio_XTD.mp4
TSV: 2025_07_24__12_22__MM1433001.TSV

Enter number of plates: 3

Detected 57 droplets in 1ul_bio_redo
Manual droplet detection for 1ul_bio_redo? (Y/N): y
Droplet count after manual correction: 60

Enter video start time (HH:MM:SS, 24h format): 12:26:36
Enter file (TSV) start time (HH:MM:SS, 24h format): 12:22:54
Enter number of skipped seconds: 30
Processing video...
Analysis complete!
```

5. Repeat for each detected folder
    a. The code will run the analysis after every folder input until there are no folders remaining

        i. There will be a pause after inputting the times for each folder. Simply wait and let the code complete its analysis
    b. As a sanity check, the code will sometimes quickly open and close a new window when downloading images.
6. The code will output a Brightness Derivative plot, a Binary Detected Droplets map, and a Detected Droplets image in the same folder that the data is stored in (ie: all data for sample A_28 will be stored in the A_28 folder)
7. The code will output a Freezing Results Excel sheet in the same folder as it. Open this file to find the freezing points of the droplets
    a. If the code successfully completes, it will print the name of the downloaded Excel file in the console

```
Saved: 2025_08_14_freezing_results.xlsx
```

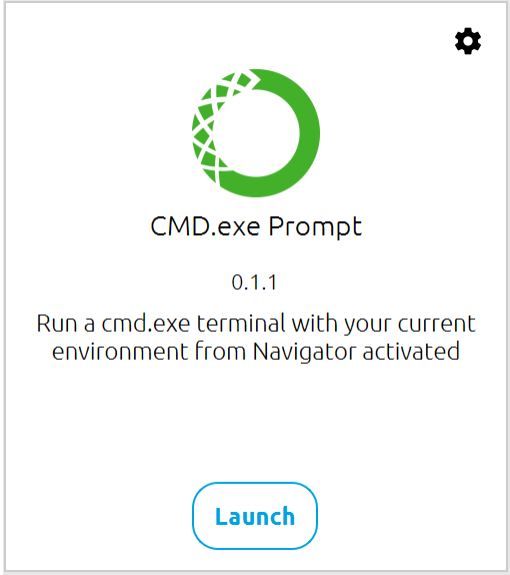
    b. Open the excel file to make sure all droplets have a corresponding freezing temperature. If some droplets a missing a temperature, try lowering the prominence parameter ("Setting Parameters" section of this document)
    c. Or, refer to *Bertram Group OneDrive > SOPs > Droplet Freezing Experiments > Droplet Freezing Data Analysis (MATLAB)_July 2024.docx* for manual analysis instructions
    d. The folder should look something like this once the code is done

Name	Date modified	Type	Size
1ul_bio	8/1/2025 10:35 AM	File folder	
1ul_water	8/1/2025 10:38 AM	File folder	
2ul_bio	8/1/2025 10:35 AM	File folder	
2ul_water	8/1/2025 10:35 AM	File folder	
freezing_analysis_1.5.py	8/14/2025 12:12 PM	PY File	15 KB
2025_08_14_freezing_results.xlsx	8/14/2025 12:29 PM	Microsoft Excel Work...	28 KB

    e. **Make sure to copy all necessary information from the Excel file before re-running code (for redoing trials or another reason) in the same folder as it will write over the old file**
    f. Close the excel file when rerunning analysis in the same folder. The xlsxwriter cannot edit spreadsheets currently open by the user.

Installing Required Libraries

1. Go to the bottom left search bar and type in “Anaconda Navigator”. Open the application.
2. In Anaconda, open “CMD.exe prompt”



3. Paste this line in the terminal and press enter
    a. “pip install opencv-python numpy pandas matplotlib scipy openpyxl”
    b. This will install all the required libraries, skipping those that have already been installed
4. This step only has to be done once; the first time before using the code

Setting Parameters

```
# === PARAMETERS ===
BRIGHTNESS_PEAK_PROMINENCE = 30  # brightness derivative threshold for freez
# lower this if droplets are detected but no freezing point is determined
MIN_RADIUS = 8  # min radius for droplet detection (pixels)
# change this if droplets aren't being detected properly
FRAME_SKIP = 0 # number of frames to skip during analysis
# default value: 0 (analyse every frame)
```

There are three adjustable parameters for the code (shown above):

- Prominence
    - Ideally, this does not need to be changed
- Radius
    - Change this based on video resolution and droplet size
- Frame Skip
    - Should always be 0 unless your video is very long and frame-perfect precision isn’t required

Common errors

1. Code detecting background noise
    a. Try to increase min_radius
    b. Increase min_radius only in increments of 1 because it is very sensitive
        i. Increase this if the code detects very small droplets outside the plate
    c. If increasing min_radius doesn't work, edit the adaptive binary thresholding
        i. Decrease both numbers (79, -5) to decrease sensitivity (LEAVE 255)
        ii. The second number (-5) is VERY sensitive, so decrease it slowly.

```
# adaptive binary thresholding
binary_image = cv2.adaptiveThreshold(
    blurred, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
    cv2.THRESH_BINARY, 79, -5
)
binary_vis = cv2.cvtColor(binary_image, cv2.COLOR_GRAY2BGR)
```

2. Code not detecting droplets
    a. Inversely to the problem above, if the code isn't detecting many droplets, decrease min_radius
    b. Try increasing the two numbers in the adaptive thresholding as described above
3. Mis-identifying or outputting blank freezing temps
    a. If the code is detecting the droplet, but doesn't output a freezing point for the droplet, check the first derivative of brightness plot.
    b. Raise or lower the "brightness_peak_prominance" value according to the peaks on that plot
        i. Code detects freezing point at the highest peak *only* if it's above the prominence threshold
4. Console outputting "no access" to Excel file
    a. The Excel file is likely open. Close the file and run the code again
    b. If the code is still having issues editing the file; close Spyder, delete the excel file (or move it out of the folder), and run the code again